



BITS Pilani

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CE F231 Fluid Mechanics

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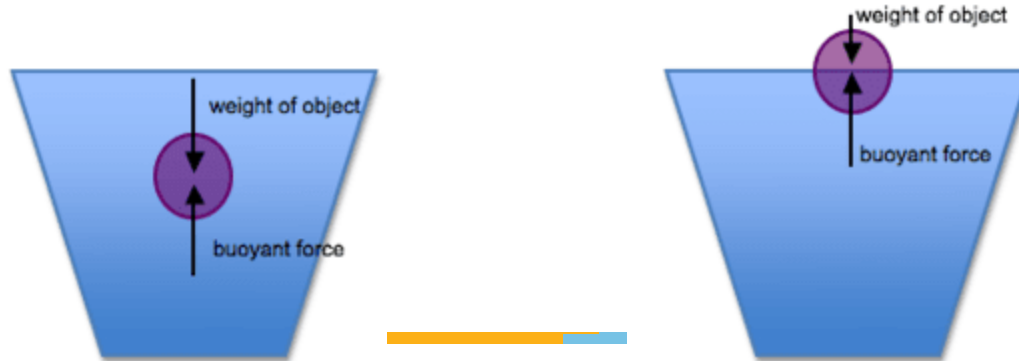


Lecture 12: Buoyancy and Floatation

Recap: Buoyancy and Archimedes Principle



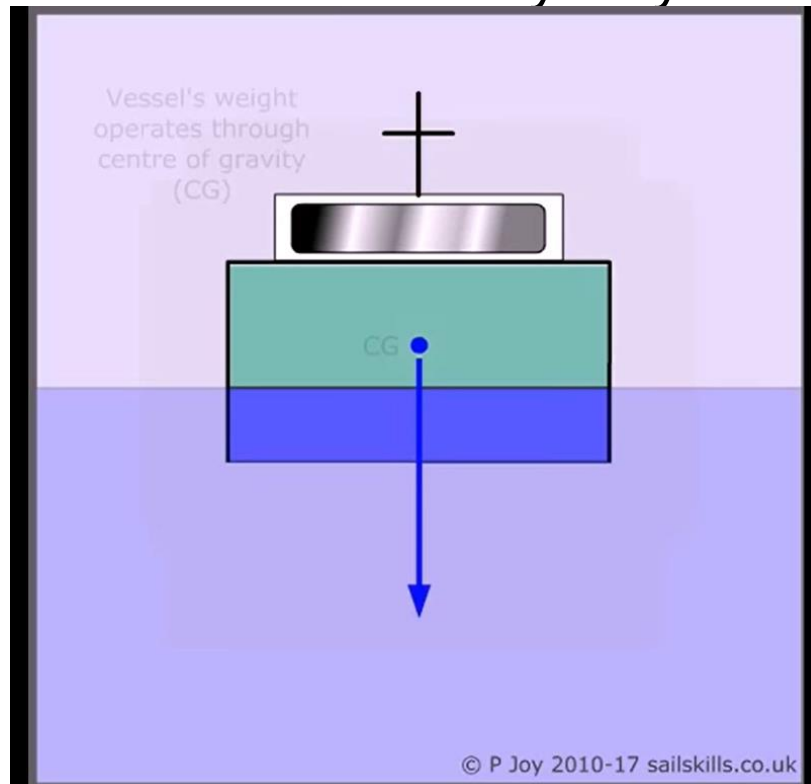
- Archimedes principle states that any body completely or partially submerged in a fluid at rest is acted upon by an upward, or buoyant force, the magnitude of which is equal to the weight of the fluid displaced by the body.
- The volume of displaced fluid is equal to the volume of an object fully immersed in a fluid or to that fraction of the volume below the surface for an object partially submerged in a liquid.



Recap: Centre of Gravity and Centre of Buoyancy



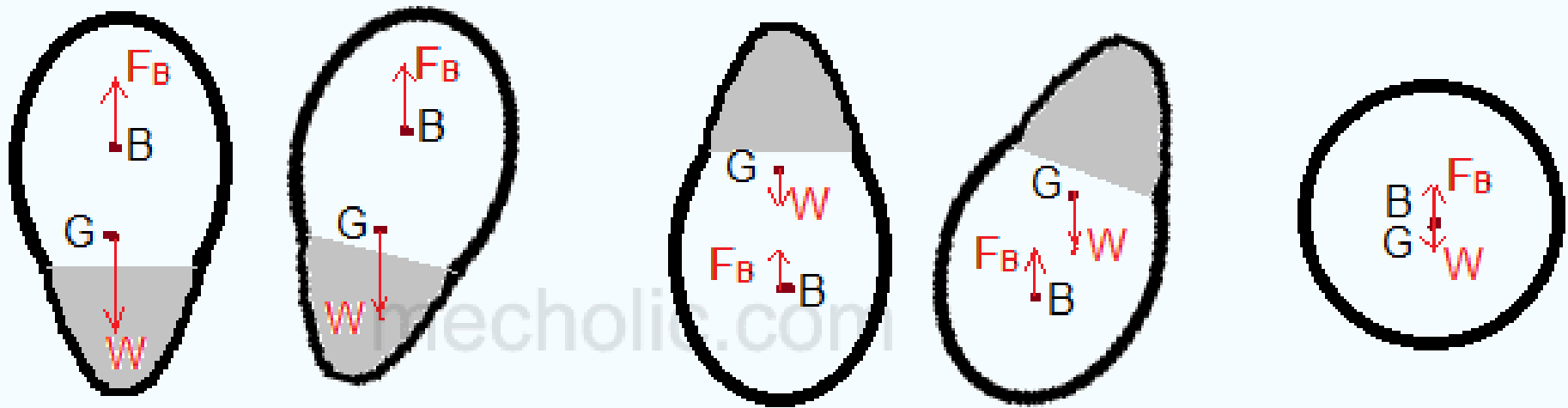
- Centre of gravity: Point through which weight of body is supposed to act
- Centre of buoyancy: Point through which buoyant force is supposed to act. Centre of buoyancy = centroid of volume of displaced fluid



<https://www.youtube.com/watch?v=AC9EULjVbo>

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Recap: Stability of Submerged Bodies

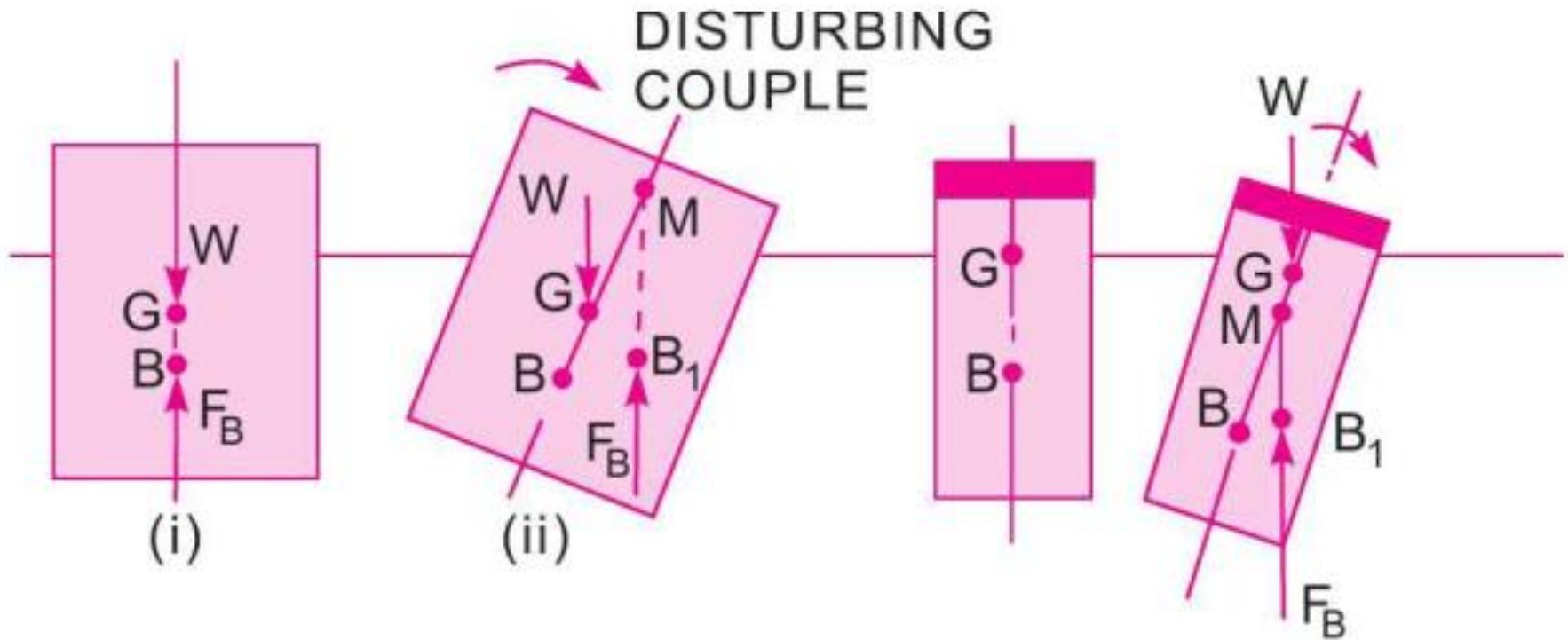


- Stable if G is below B
- Unstable if G is above B
- Neutral if G and B coincides

Stable or unstable → classified based on whether the body is able or unable to regain initial position after disturbing/tilting

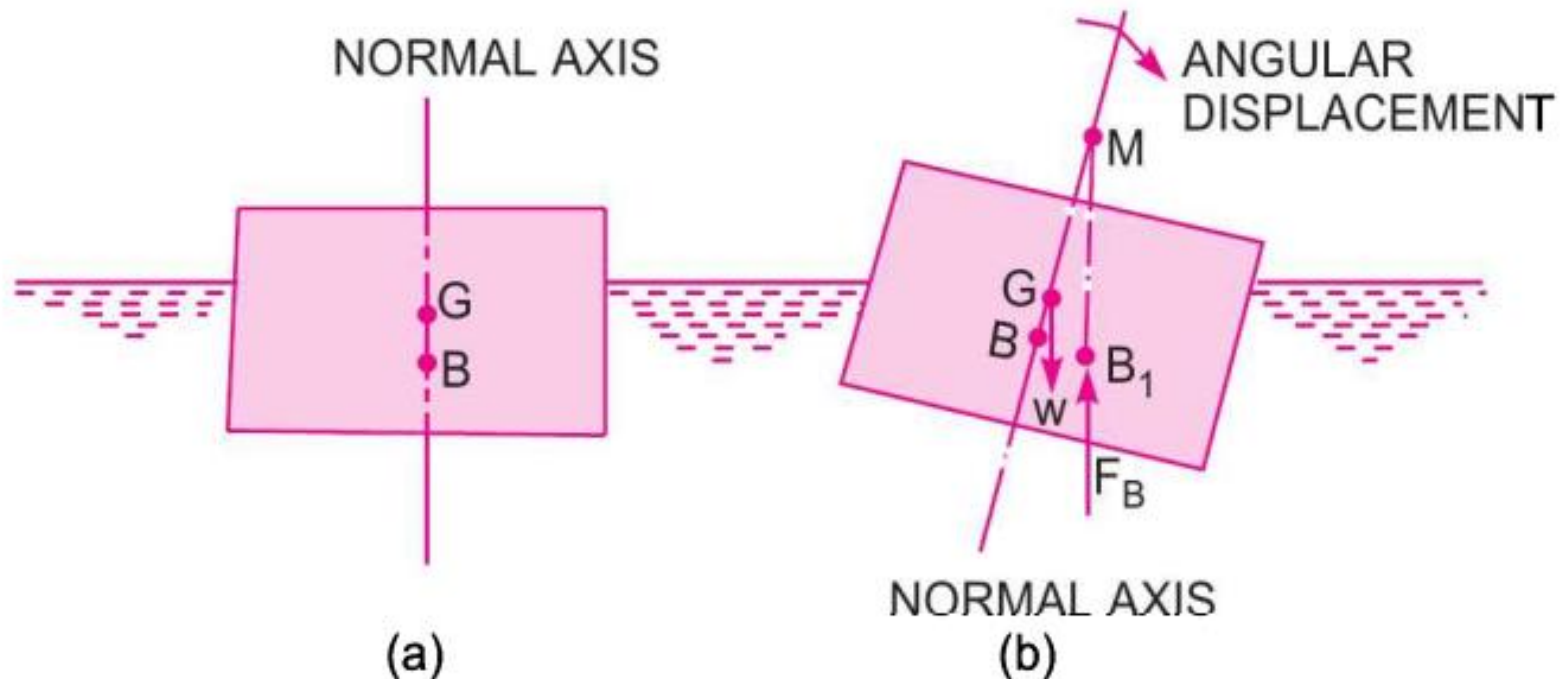
Recap: Stability of Floating Bodies

- Unlike submerged bodies, we cannot comment on stability of floating bodies solely based on relative locations of G and B

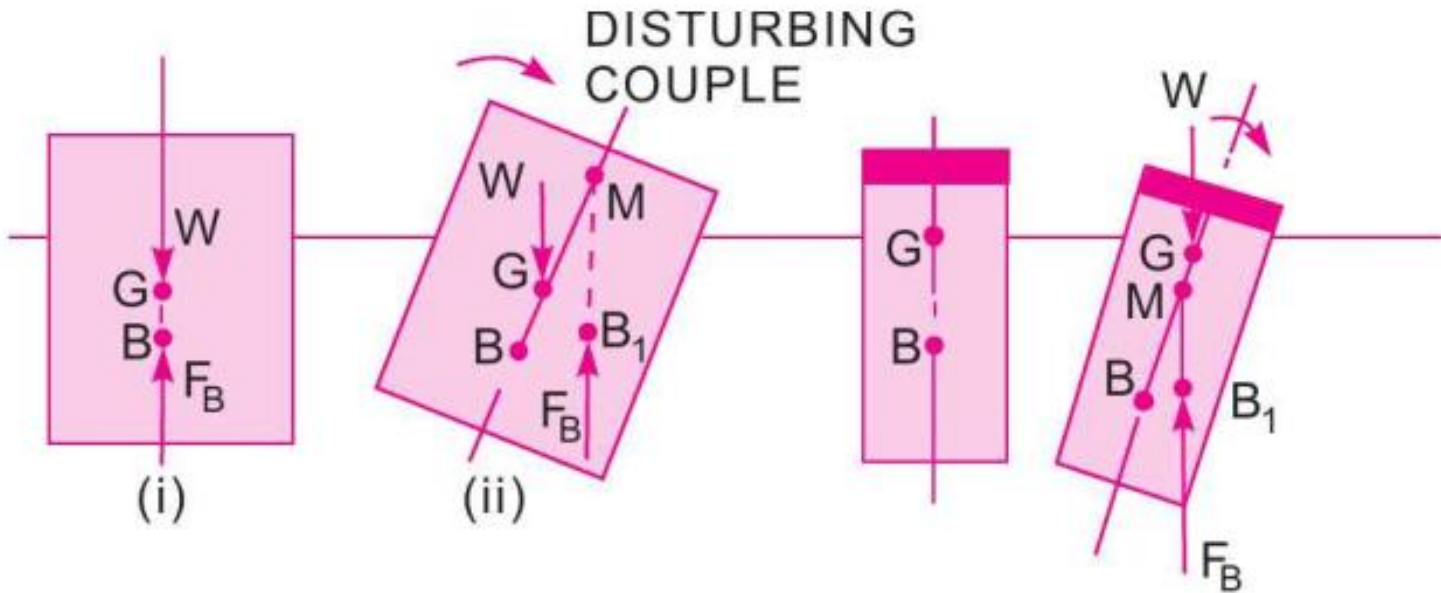


Recap: Metacentre

- After imparting a small angular displacement, the line of action of buoyant force will intersect the original normal axis at point M – metacentre

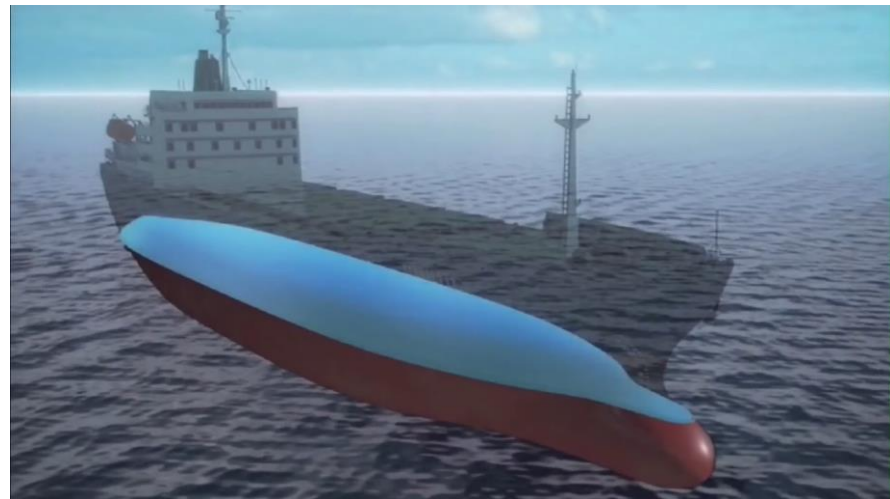
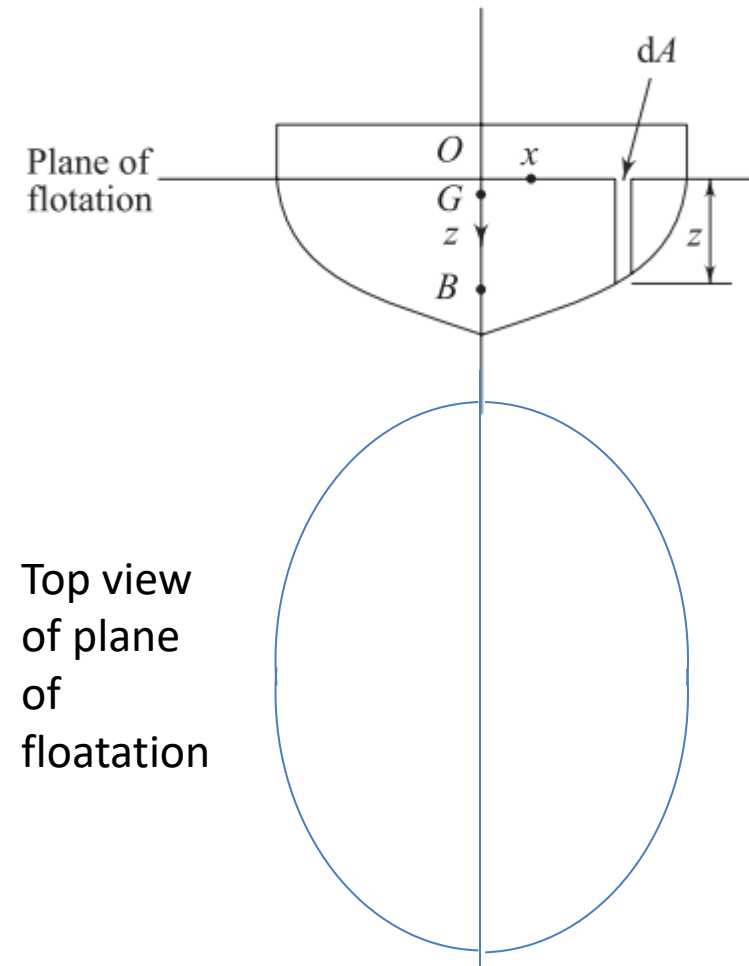


Recap: Metacentric Height and Stability of Floating Bodies



- Metacentric height GM = distance b/w CG and M
- $GM = BM - BG$
 - $GM \rightarrow +ve \rightarrow M$ is above $G \rightarrow$ stable
 - $GM \rightarrow -ve \rightarrow M$ is below $G \rightarrow$ unstable
 - $GM \rightarrow 0 \rightarrow G$ and M coincides $\rightarrow$ neutral

Analytical Method to Estimate Metacentric Height

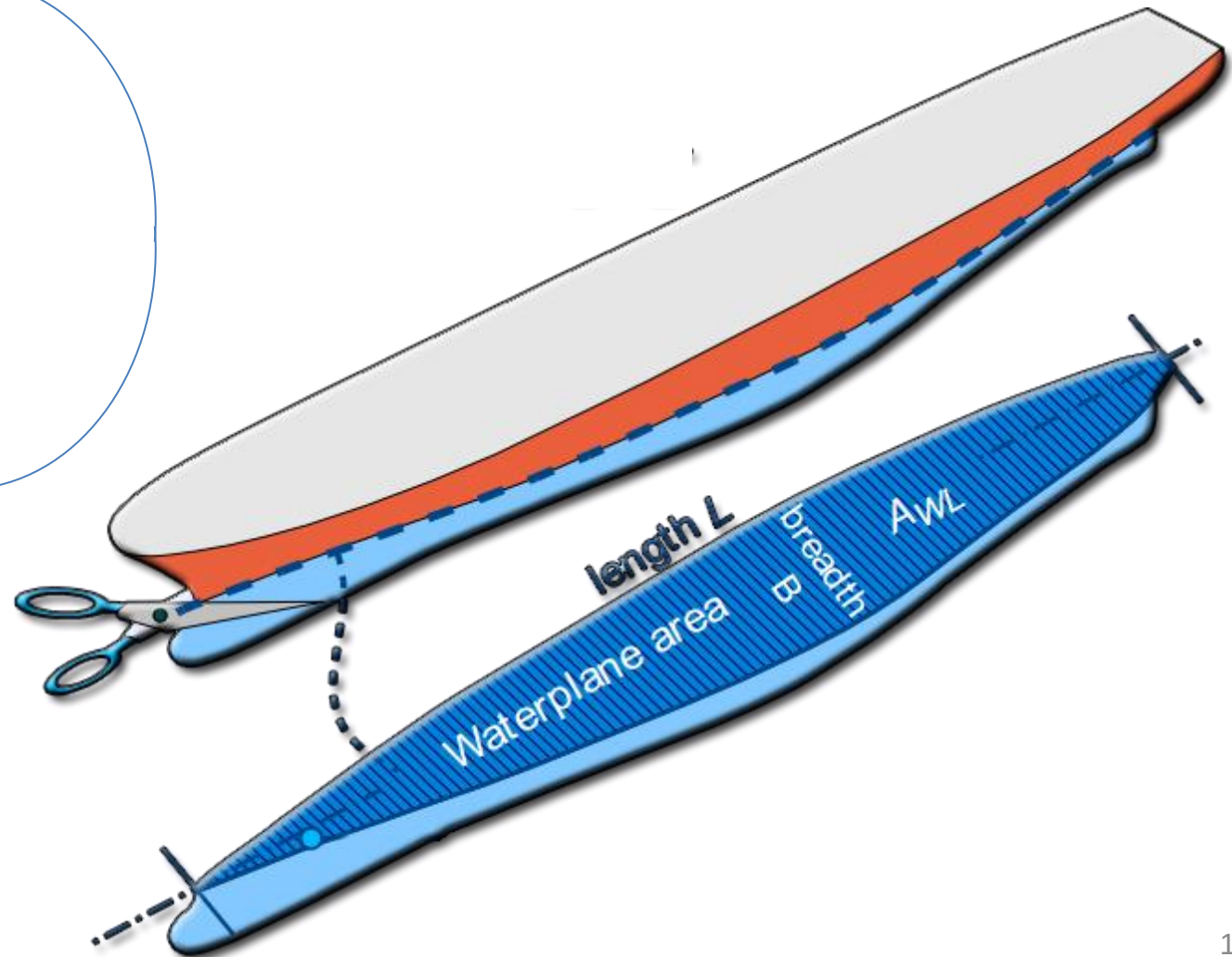
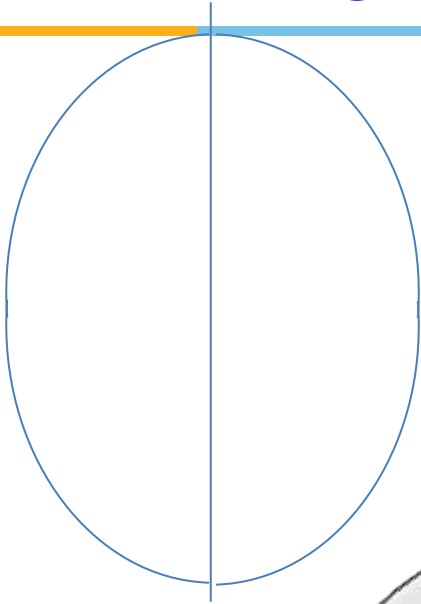


Plane of floatation: Horizontal intersection of water plane and vessel

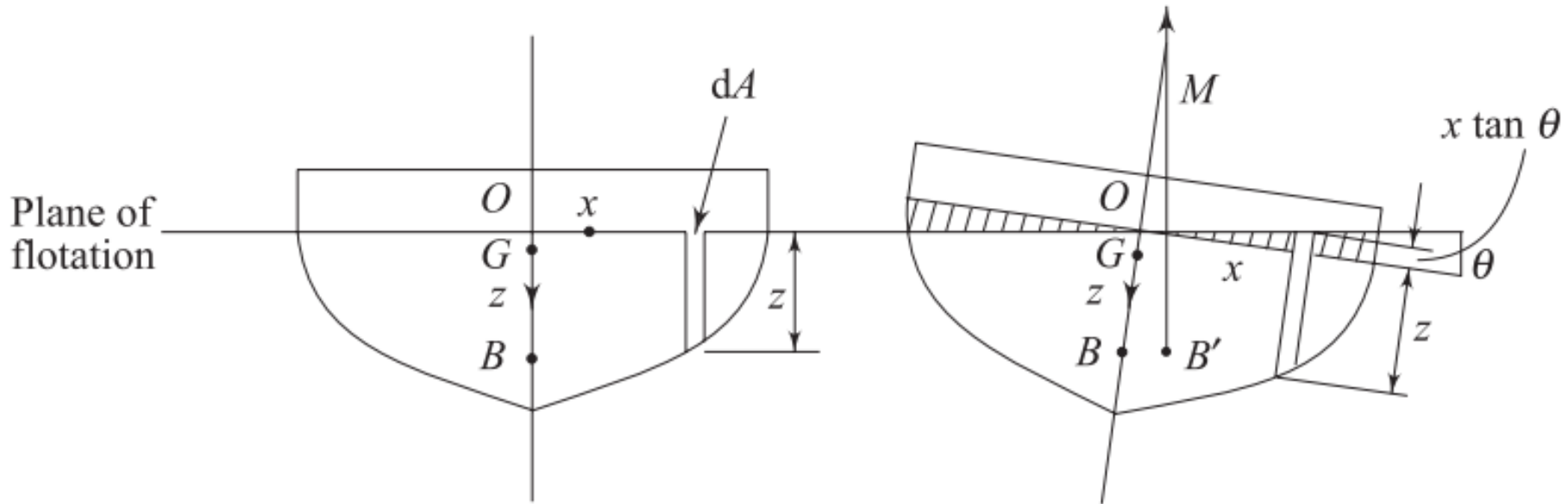
Analytical Method to Estimate Metacentric Height



Top view of
plane of
floatation
(waterplane
area)



Analytical Method to Estimate Metacentric Height



- Aim: To analytically determine an expression for BM and then GM
- BM and BB' and related $\rightarrow \sin \theta = BB'/BM = \tan \theta$ (for small θ)
- BB' is the difference between x coordinates of center of buoyancy after and before the tilt

$$GM = \frac{I}{\nabla} - BG$$

Pitch and Roll



Pitch → Motion about transverse axis

Roll → Motion about longitudinal axis

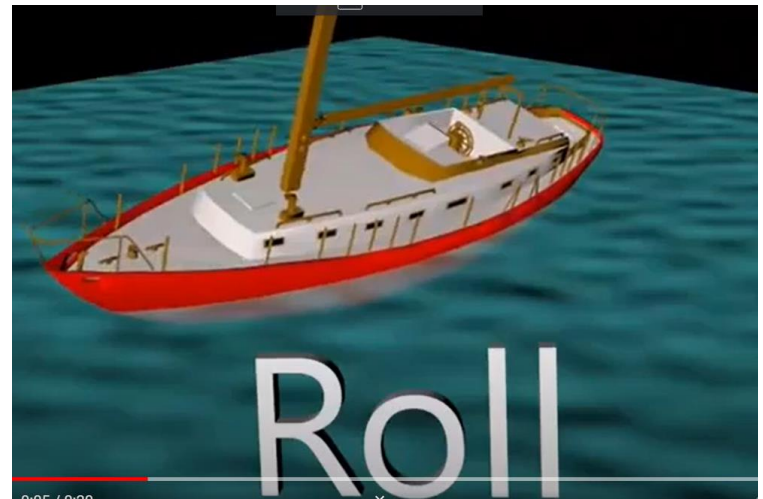
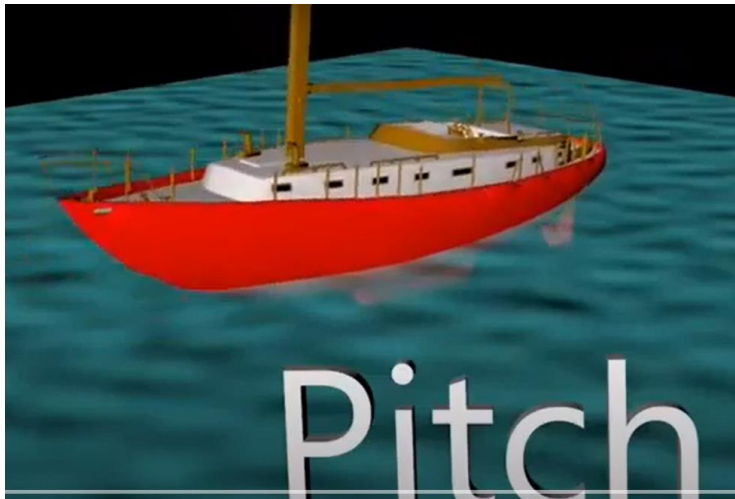
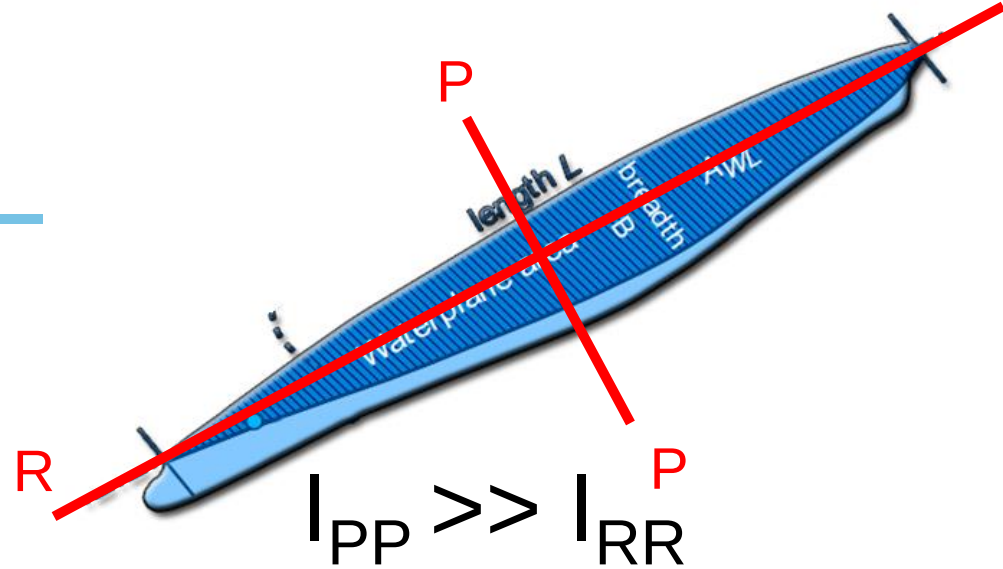
Pitch and Roll

$$GM = \frac{I}{\Delta} - BG$$

GM for Pitch

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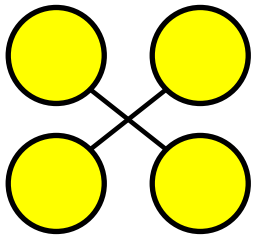
GM for Roll



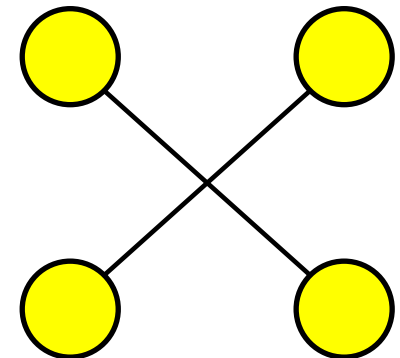
Question



Why the left configuration is unstable, but the right configuration is stable?

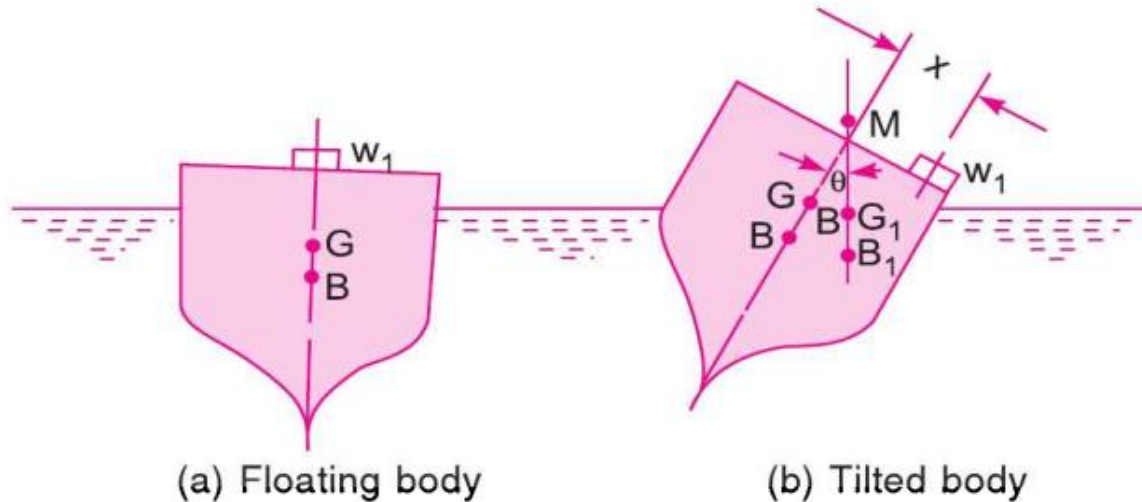


$$GM = \frac{I}{\nabla} - BG$$



Assume combined weight of buoyancy tanks and connectors are the same for both scenarios

Experimental Determination of Metacentric Height



Let W = Weight of vessel including w_1
 G = Centre of gravity of the vessel
 B = Centre of buoyancy of the vessel

The moment due to change of $G = GG_1 \times W = W \times GM \tan \theta$

The moment due to movement of $w_1 = w_1 \times x$

$\therefore$

$$w_1 x = WGM \tan \theta$$

Hence

$$GM = \frac{w_1 x}{W \tan \theta}$$

Time Period of Oscillation

$$T = 2\pi \sqrt{\frac{K^2}{GM \cdot g}} \qquad I_{Y-Y} = \frac{W}{g} K^2$$

*T = time period; K = radius of gyration; I = mass moment of inertia;
W = weight; g = acc due to gravity*

- Higher value of GM → Greater stability of vessel → Lower time period of oscillation → Higher frequency of oscillation → Higher discomfort for passengers
- Lower value of GM → Lower is the stability of vessel → Higher time period of oscillation → Lower frequency of oscillation → Lesser discomfort for passengers



THANK YOU